

Advanced Materials „Soft Matter“

**Institut für Organische Chemie und
Makromolekulare Chemie
der Heinrich-Heine-Universität Düsseldorf**

Ausgabe SoSo 2023

Contents

Contents	1
1 Hydrogels.....	2
2 Synthesis of Microgels.....	5
3 Poly(ethylene glycol) (PEG).....	6
4 Introduction on light microscopy,	8
4.1. Optical resolution	10
4.2. Enhancing the contrast of optical microscopy images: phase contrast imaging.....	11
4.3. Fluorescence Microscopy	13
5 Introduction to atomic force microscopy (AFM)	15
5.1. The technique: How to sense interaction forces in molecular-scale? ..	16
5.2. Forces in AFM	18
5.3. Contact Mode AFM	20
5.4. Vibrating mode AFM: Non-contact and intermittent contact	21
5.5. Intermittent contact mode (also termed tapping mode)	22
5.6. Piezoelectric scanners and scanning artifacts.....	23
5.7. Scanning artifacts due to tip wear and convolution.....	24
6 Introduction to AFM Force spectroscopy.....	26
6.1. Simple force curves	27
6.2. More complex interactions	28
6.3. Applications	29
7 Practical Course	30
7.1. Synthesis of non-Covalently Cross-linked Polymer Gel	31
7.2. Synthesis of Chemically Cross-linked Gel-Building Polymer	33
7.3. Synthesis of poly-(N-isopropylacrylamide) microgels via precipitation polymerization	35

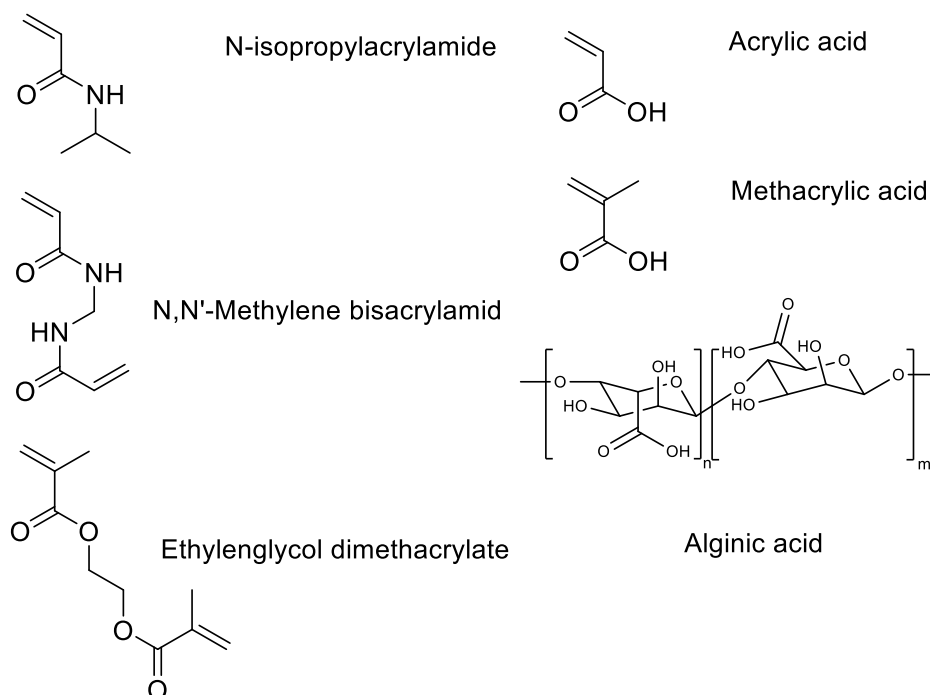
1 Hydrogels¹

Hydrogels are water-swollen hydrophilic polymeric materials that maintain a distinct three-dimensional structure. The high-water content of the material contributes to their biocompatibility. Furthermore, their hydration results in tissue-like mechanical properties and makes them highly suitable for *in vivo* applications. Hydrogel materials are studied increasingly for applications in biological sensing, drug delivery, and tissue regeneration.

Hydrogel networks can be composed of either homopolymers or copolymers and are insoluble due to the presence of chemical or physical cross-links. The latter are also called 'reversible', or 'physical' gels that are held together by molecular entanglements and/or secondary forces including ionic, H-bonding or hydrophobic forces. Physical hydrogels can reversibly degrade into the corresponding precursors upon external stimuli. Hydrogels that are covalently cross-linked are called 'permanent' or 'chemical' gels. Chemical hydrogels may be generated by cross-linking of water-soluble polymers in the presence of cross-linkers. In the cross-linked state, hydrogels reach an equilibrium swelling level in aqueous solution that depends mainly on the cross-link density.

Hydrogels can be obtained as macroscopic networks or in smaller dimensions as so-called **microgels**. Microgels are usually spherical, have sizes from 100 nm to 100 µm and thus large surface area. In the context of biomaterials and biotechnology, the microgel surface is often modified by bioconjugation in order to introduce ligands, e.g. carbohydrates or proteins that specifically bind to targeted sites. Scheme 1 shows some examples of the most common mono- and macromonomers for the synthesis of chemical gels.

¹ Daniel Pussak, Synthesis and Functionalization of Colloidal Probes based on Poly(ethylene glycol) as Carbohydrate Biosensors, PhD Thesis 2014, Department of biology, Chemistry and Pharmacy of Freie Universität Berlin, Germany



Scheme 1: Structures of common monomers and macro monomers used to prepare microgels

Some microgels are able to respond to chemical and physical stimuli, like temperature, pH changes, ionic strength or light, changing e.g. their shape, aggregation behavior, degradation or swelling/collapsing.[4] One of the most important properties of hydrogels is their swelling behavior. This, and specifically the swelling degree, depends on the polymer-polymer and polymer-water interaction. When the microgels are dispersed in a good solvent e.g. H_2O , the polymer chains extend to increase the polymer-water interaction thereby swelling the gel. Gels undergo a transition from a swollen state in a good solvent to collapsed state in a bad solvent, when the polymer-polymer interaction becomes dominant. Microgels prepared of hydrophilic polymers will therefore swell in water and collapse in organic solvents with low dipole moment, whereas microgels composed of non-polar polymers such as polystyrene swell in non-polar solvents such as toluene. For hydrogels made of polyelectrolytes such as methacrylic acid, a pH dependent swelling can be observed. For example, methacrylic acid as well as acrylic acid microgels aggregate and collapse at neutral pH and re-swell at basic conditions. Here, the deprotonated acid groups show electrostatic repulsion under basic conditions leading to a swelling of the microgel.

The most widely studied so-called 'smart' microgel is made of poly(*N*-isopropylacrylamide (PNIPAm). PNIPAm is a thermoresponsive polymer with lower critical solution temperature (LCST) around $32\text{--}33^\circ\text{C}$. PNIPAm microgels exist in swollen state at temperatures below 32°C , where PNIPAm associates water by hydrogen bonding. When the temperature is increased, water is released due to entropic effects, the polymer-polymer interactions are favored and the polymer reorganizes to a collapsed state by dehydration of the polymer.

Microgels present many other attractive features such as their colloidal stability and their chemical versatility. They can be synthesized in a variety of sizes, morphologies, chemical compositions and with different functionalities.

2 Synthesis of Microgels²

The preparation of microgels can be divided into two groups: physical cross-linking or chemical cross-linking. The physical cross-linking method involves hydrophobic or electrostatic interactions to cross-link polymer chains into particles. This is a common procedure for the preparation of microgels from polyelectrolytes such as alginate or chitosan.

Chemical cross-links can be achieved for example *via* free radical polymerization in the presence of multifunctional cross-linkers, e.g. acrylic monomers can be cross-linked by bisacrylamide. Precipitation, suspension or emulsion polymerization are the most commonly applied routes towards microgel preparation. In a precipitation polymerization, the monomer and the initiator are soluble in a solvent and upon cross-linking of the monomer, the polymer collapses and precipitates forming a separate phase (dispersion) in the solvent. As the reaction proceeds, flocculation and aggregation of the initially formed nuclei take place. In case of continuous formation of new nuclei, a polydisperse precipitate with a broad spectrum of irregularly shaped particles is formed.

Monodisperse microgels can result, if all nuclei are formed in a time frame that is much shorter than the time frame of the nuclei growth process. This technique has been used for the synthesis of PNIPAm and other thermoresponsive microgels. In suspension polymerization the initiator and the monomer are stabilized by stirring and a suitable droplet stabilizer suspended in the medium in the form of microdroplets. The polymerization is then initiated and the monomer microdroplets are converted directly to the corresponding polymer microgels usually with the same size as the microdroplets. In emulsion polymerization, the polymerization takes place in monomer microdroplets that form spontaneously in the continuous phase where the initiator and free surfactant is placed to stabilize the monomer emulsion. Emulsion polymerization usually leads to submicron sized particles and narrow particle size distributions.

The size of the microgel particles in the aforementioned techniques is either limited by the nucleation and growth kinetics (e.g. precipitation) or thermodynamics (emulsion polymerization of the polymer phase). Microfluidic techniques can be used for the synthesis of monodisperse microparticles of virtually arbitrary sizes. To this end, microfluidic devices are used to prepare emulsions of the monomer solution in the size range of 1-50 μm followed by physical or chemical cross-linking.

² Daniel Pussak, Synthesis and Functionalization of Colloidal Probes based on Poly(ethylene glycol) as Carbohydrate Biosensors, PhD Thesis 2014, Department of biology, Chemistry and Pharmacy of Freie Universität Berlin, Germany

3 Poly(ethylene glycol) (PEG)³

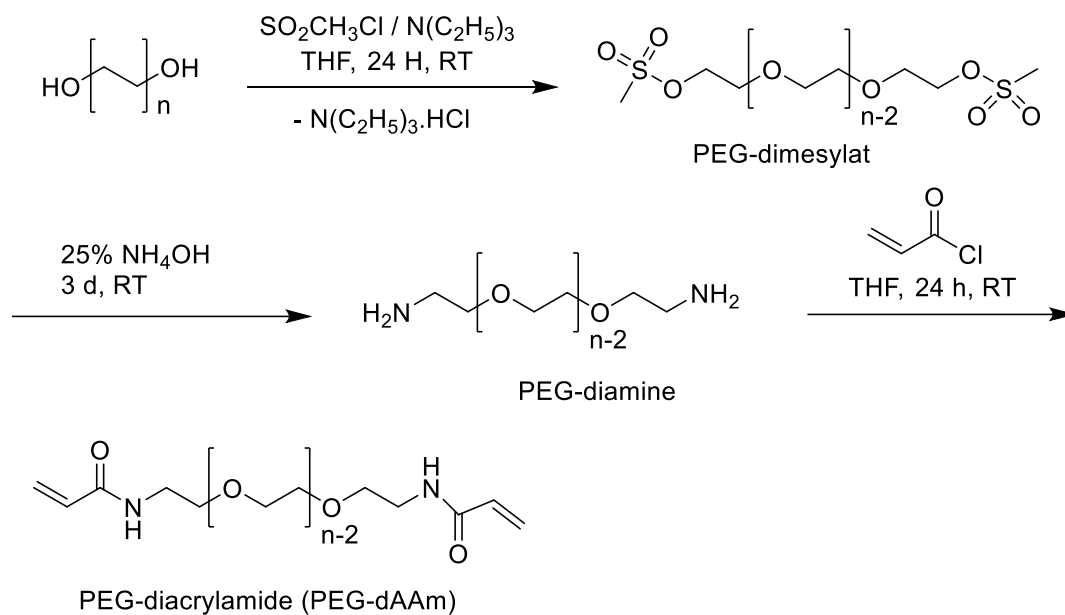
PEG is a polyether and synthetically derived from ethylene oxide via anionic polymerization. The initiation of the reaction with hydroxide ions leads to PEG chains with hydroxyl groups on both ends. Different terminal groups can be obtained when the type of initiator is changed. The PEG geometry is not limited to linear chains. Other geometries have been prepared such as branched or star PEG. It is believed that PEG's inertness comes from the hydrophilicity, the inability to donate hydrogen bonds, and conformational flexibility of the backbone that makes it difficult for proteins to adhere on e.g. PEGylated surfaces. The term PEG is used for polyethers of molecular weights below 20000 g/mol while poly(ethylene oxide) (PEO) refers to larger polymers. PEG is commercially available and with molecular weights below 1000 g/mol it is a colorless viscous liquid while larger PEG's are white solids at room temperature. The melting point of PEG increases with molecular weight and reaches a plateau at 67°C. PEG is also a thermoresponsive polymer that phase separates in an aqueous solution at temperatures higher than 100°C.

In most applications PEG is conjugated to drugs or surfaces (PEGylation) to protect against degradation or coagulation. Additionally, PEG is used in hydrogels by cross-linking PEG-macromonomers. A variety of PEG-macromonomers are available either by choosing the appropriate initiator during polymerization or by post-functionalization of the terminal end-groups of PEG, most often the functionalization of terminal hydroxyl groups. In this way functional groups such as carboxylates, amines, thiols or azides can be introduced. All these groups can be used to prepare PEG chains with cross-linkable end-groups. By introducing acrylates or acrylamides as end-groups no additional cross-linker is needed for the preparation of hydrogels, thus resulting in 100% PEG materials. The synthesis of PEG-diacrylamides (PEG-dAAm) is shown in Scheme 2.

As PEG does not have any further functional groups on the backbone, different strategies are available to achieve functional PEG microparticles. A second monomer bearing vinyl groups can be co-polymerized with PEG-dAAm to obtain additional functionalities.

However, this method does not give pure PEG hydrogels anymore but always leads to changes of the particle properties such as the cross-linking density and the mechanical stability. Alternatively, ready-made PEG microgels can be functionalized in a polymer analogue fashion.

³ Daniel Pussak, Synthesis and Functionalization of Colloidal Probes based on Poly(ethylene glycol) as Carbohydrate Biosensors, PhD Thesis 2014, Department of biology, Chemistry and Pharmacy of freie Universität Berlin, Germany



Scheme 2: Synthesis of end-group functionalized PEG for SCP preparation

4 Introduction on light microscopy⁴,

A light microscope is an optical instrument that uses light to produce a magnified image of an object that is projected onto a detector – the eye or a camera. A typical light microscope consists of two main lenses, the objective lens and the ocular lens working together to give the final magnification $M_{\text{final}} = M_{\text{obj}} \times M_{\text{oc}}$, where M_{obj} is the magnification of the objective and M_{oc} is the magnification of the ocular. (see Figure 1)

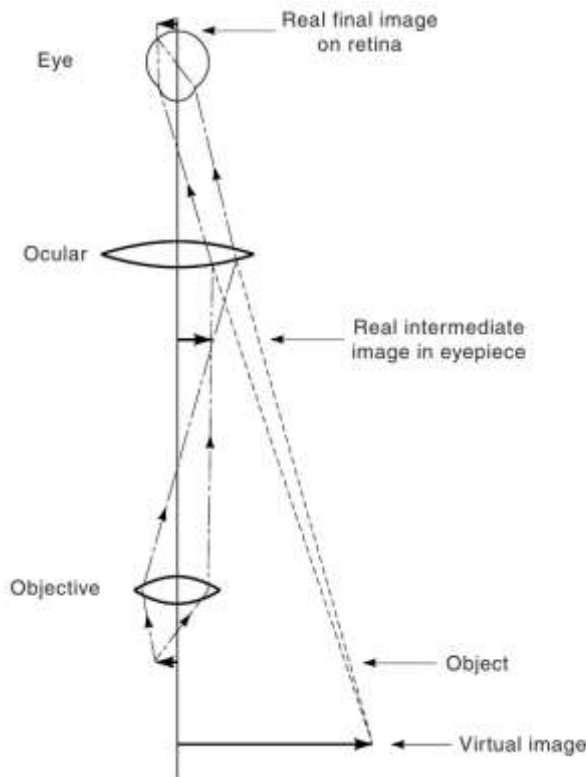


Figure 1: Perception of a magnified virtual image of a specimen in the microscope. The objective lens forms a magnified image of the object (called the real intermediate image) in or near the eyepiece; the intermediate image is examined by the eyepiece and eye, which together form a real image on the retina. A real image can be viewed on a screen or exposed on a sheet of film, whereas a virtual image cannot

Two microscope components are of critical importance in forming the image: the objective lens, which collects light, diffracted by the specimen and forms a magnified real image at the real intermediate image plane near the eyepieces or oculars, and the condenser lens (Figure 2). Both the objective and condenser contain multiple lens elements to maximize performance and minimize optical artifacts. Other components like tube and eye-pieces, the lamp collector and lamp socket, filters, apertures, polarizers, retarders, and the microscope stage and stand with coarse

⁴ Douglas B. Murphy, Fundamentals of Light Microscopy and Electronic Imaging, Wiley, ISBN-13: 978-0471692140

and fine focus dials serve to acquire the images, illuminate the specimen or allow for specialized contrasting techniques.

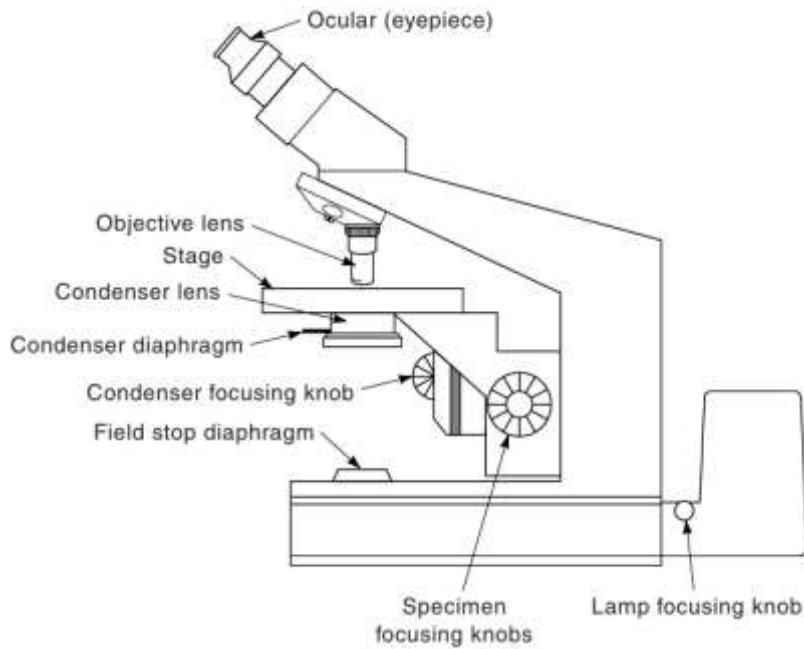


Figure 2: A typical upright light microscope. Note the locations of the specimen focus dials, the condenser focus dial, and the focus dial of the collector lens on the lamp housing. Also note the positions of two variable iris diaphragms: the field stop diaphragm near the illuminator, and the condenser diaphragm at the front aperture of the condenser. Each has an optimum setting in the properly adjusted microscope.

4.1. Optical resolution

Magnification alone is not enough: The resolution determines what we see. This is explained in Figure 3.

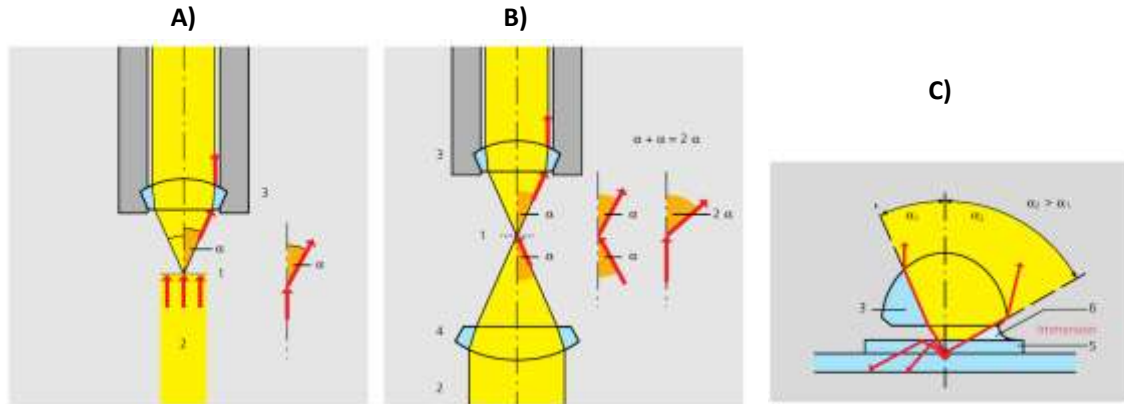


Figure 3: A) If small objects (1) are viewed through the microscope, the light (2) incident from these objects is deflected (diffracted) from the original direction. The smaller the objects are, the more pronounced the deflection will be. To obtain sharp images of small structures, the objective (3) in the microscope must collect as much of this diffracted light as possible. This works particularly well if the objective covers a large solid angle. The term aperture (“opening”) describes this property.

B) One way of increasing the optical resolving power is to choose a large angle of the ray cone also on the illumination side. This enables the objective to gather even light rays displaying very pronounced diffraction. Such strongly deflected rays originate from very fine structures. A condenser (4) with the same numerical aperture as the objective is now used for illumination.

C) Another possibility is to use immersion liquids (6) between the front lens of the objective (3) and the cover slip (5). Good results have been obtained with a certain oil with the refractive index matched to the refractive index of glass. All reflections on the path from the object to the objective are eliminated in this way. If this trick were not used, reflection would always cause a loss of light in the cover slip or on the front lens in the case of large angles (Fig. 4.3 – left half). The useful aperture of the objective and the resolving power would be reduced by this reflection.

To enable the resolution of objectives, the “numerical aperture” (N.A.) has been defined: this is a measure of the solid covered by an objective. The following definition is made: $N.A. = n \times \sin \alpha$, where α is the half opening angle of the objective and n is the refractive index of the immersion medium ($n = 1$ for air, $n = 1.51$ for oil or glass). But what does “resolution” actually mean? The limit up to which two small objects are still seen separately is used as a measure of the resolving power of a microscope. A certain distance d_0 exists where this limit is reached which is due to the unavoidable diffraction pattern light generates when illumination small objects. The higher the aperture of the objective (N.A. Obj.) and of the condenser (N.A. cond.), the smaller d_0 will be. A short wavelength λ is also beneficial for the resolving power $d = \lambda / 2 N.A.$ Therefore the theoretical resolution limit of an optical microscope would be on the order of several 100 nm. However, this would mean that all optical components work at their theoretical limit. In practice the resolution is limited far more often due to sample preparation and low contrast, see following section.

4.2. Enhancing the contrast of optical microscopy images: phase contrast imaging

In optics, contrast is defined as the difference in light intensity between the objects in an image. Owing to the poor contrast of typical biological samples in liquid, e.g. proteinic structures in cells, they are poorly visible in common bright-field microscopy. However, although these structures are transparent, they can induce a phase shift to a fraction of light that passes them. Unfortunately, the eye and cameras cannot detect differences in the phase. Phase contrast microscopy aims to translate phase changes to large variations in the amplitude. The physical reason for phase shifts in the specimen are optical density gradients which lead to diffraction of light in the sample. The diffracted fraction of the light (D) will be scattered in many directions and has its phase retarded by $\lambda/4$ with respect to the undeviated light (U), which passes through the sample without interacting with it. Both the U and D light are collected by the microscope optics, will undergo interference, and generate the resulting light (R). Typically only a minority of incident light is diffracted by proteinic objects in aqueous solution. Also, the objects are only detected if the amplitudes of R and U are significantly different. Figure 4A shows that the amplitudes differences of R and U depend on the phase difference ϕ and the amplitude of D. Phase contrast microscopy manages to optimize intensity differences between R and U. This is achieved by a) separating the undeviated light U from the most of the diffracted light D so that they occupy different locations at the back aperture of the objective lens, where they are recombined shortly after (Figure 4B), and b) advance the phase and reduce the amplitude of the U light, in order to maximize differences in amplitude between the object and background in the image plane. Figure 4B and C shows that with a condenser annulus and a phase plate these requirements can be fulfilled. The condenser annulus, a black plate with a transparent annulus illuminates the specimen with a ring of light. In Köhler Illumination the U light then forms a bright image at the back aperture of the objective, while D light is across the entire back aperture. The amount of D light depends on the density and refractive index of light-scattering objects in the specimen. To strongly alter the phase and amplitude of U with respect to the R a phase plate is mounted in the phase contrast objectives, located at or near its back focal plane. The phase plate retards the U light with respect to the D light by $\lambda/2$ to enhance destructive interference. In addition it attenuates the U light by a semitransparent coating to about 20% intensity to match the intensities of U and D light (Figure 4C). Optimizing the phase relations and intensities of U and D for maximum destructive interference results in an increased intensity difference between the R-light and U light. As eyes and cameras interpret contrast as intensity difference, objects with just different optical densities appear with increased contrast.

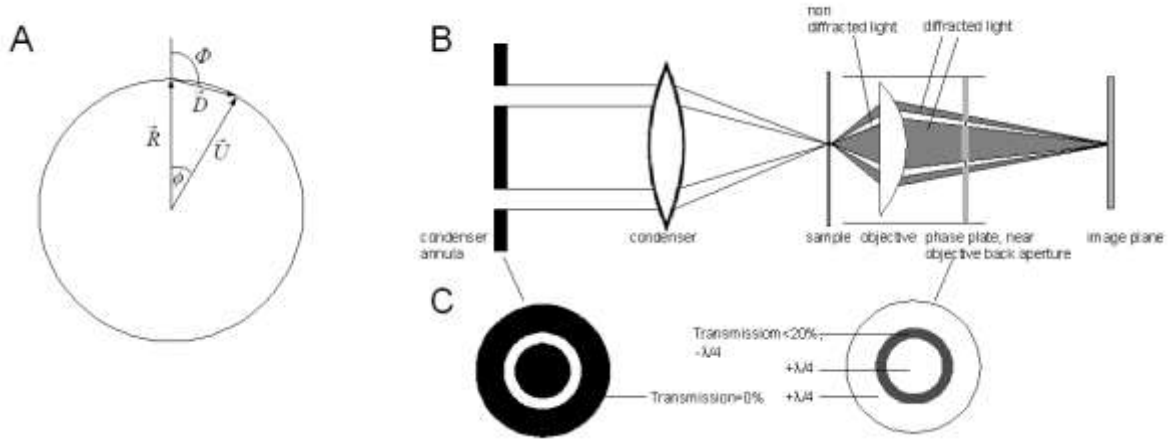


Figure 4: Phase contrast optics: A) Phasor diagram, the electric of R is shown as the vector sum of the field vectors of D and U . ϕ represents the amount of phase displacement. B) Path of the non diffracted and diffracted beams in a phase contrast microscope. The annular aperture in the front focal plane of the condenser generates a hollow cone of light that illuminates the specimen and continues (approximately) as an inverted cone that is intercepted by a phase plate at the back aperture of the objective lens. C) condenser annuli and phase plate

4.3. Fluorescence Microscopy

Instead of using the alterations of light by absorption or phase differences for visualization, in fluorescence microscopy the light emitted by labeled objects is collected exclusively. This has several advantages over conventional types of illumination. Most importantly, fluorescence microscopy allows, visualization of specific molecules that emit light in presence of excitatory light. For this reason fluorescence microscopy has become a versatile technique that allows for special measurements like binding constants and diffusion coefficients using FRAP (fluorescence recovery after photobleaching). Also observation of interaction states and reaction mechanisms are possible with fluorescence energy transfer measurements (FRET). TIRF (total internal reflection fluorescence) and laser scanning techniques allow studying dynamics of single labeled molecules in very thin focal planes.

Compared to other types of contrast enhancing techniques, the process of imaging and image interpretation is often more straight-forward in fluorescence microscopy. However, labeling the specimen with fluorescent dyes is critical, and has become a science in its own right. The physical basis of fluorescence microscopy is the excitation of the fluorophores and subsequent excited state relaxation leads to photonemission (Figure 5A). Electrons in the ground state S_0 (highest occupied molecular orbital, HOMO) are stimulated by the excitatory light to the first excited state S_1 (lowest unoccupied molecular orbital, LUMO). Upon electron relaxation to S_0 , light is emitted, with a red-shift (Stokes shift) with respect to the excitatory light, which is assigned to two different reasons. Firstly, only vertical electron transitions are allowed and secondly, relaxation of the fluorophore's vibrational levels in the S_1 state (Franck Condon principle). In fluorescence microscopes the excitation light needs to be isolated from the emitted light using a set of interference filters, so it is advantageous to use dyes with large Stokes shifts.

Construction-wise a fluorescent microscope shows a number of important characteristics. It has a very intense light source, such as a mercury arc lamp, because only a small fraction of light is suitable for sample excitation. To increase the contrast all fluorescent microscopes are built for epi-illumination, i.e. the objective lens functions both as the condenser, illuminating the specimen, and as the objective lens, collecting the emitted light. This configuration is advantageous because it avoids the strong excitatory beam being directed to the detector (Figure 5B). The fluorescence filter set contains three filter sets that further separate the excitatory light from the emitted light: the excitation filter, the dichroic mirror and the emission filter. Finally the fluorescence objectives are optimized as well; they contain low-fluorescent glass and have a high numerical aperture.

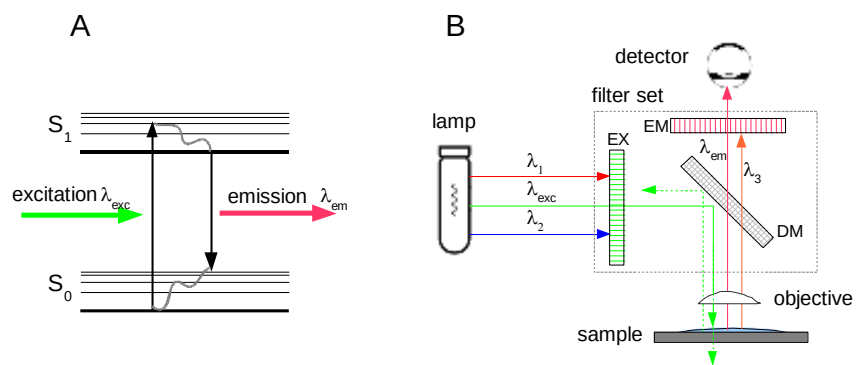


Figure 5: Scheme of the electronic (thick lines) and vibrational (thin lines) energy states of a chromophore. B: Scheme of a fluorescent microscope in epi-illumination. The filter set comprises of an excitation filter (EX), an Emission filter (EM) and the dichroic mirror (DM).

5 Introduction to atomic force microscopy (AFM)^{5,6}

An atomic force microscope can analyze the surface topography (structure), mechanical properties, as well as chemical and physical interaction forces between surfaces. Importantly, an atomic force microscope works down to molecular resolution. A rudimentary analogy between an optical microscope and an AFM would be that the optical microscope uses the “seeing” sense whereas the AFM uses the “feeling” sense in order to depict a specimen. More precisely, AFM investigations are performed by scanning the sample with a silicon needle of close to atomic sharpness, the AFM tip. While scanning the AFM tip along a surface it experiences forces due to the interaction with the sample. These interaction forces can be related to surface topography, mechanics and even chemical composition of the sample. In the following the theories behind the method are explained, i.e. nature of the interaction forces, force detection techniques by AFM cantilevers, as well as piezo scanning with ångström precision, mechanical measurements, etc.

⁵ Colton, Engel, Frommer, Gaub, Gewirth, Guckenberger, Heckl, Parkinson, Rabe Procedures in Scanning Probe Microscopies. Wiley, 1997. ISBN: 047195912

⁶ H.-J. Butt, B. Cappella, M. Kappl; Surface Science Reports 59, 1 (2005).

5.1. The technique: How to sense interaction forces in molecular-scale?

Atomic Force Microscopy (AFM), known also as Scanning Force Microscopy (SFM), was invented in 1986 by Binnig and co-workers [4]. The physical quantity that is measured with this apparatus is the force between a sharp conical tip and a sample surface while scanning the tip across the sample. This allows investigations to be performed on a wide range of materials including polymers and hydrogels. AFM tips scanning the sample surface are very sharp, with a terminal radius lower than 10 nm for the most common types of AFM tips. The tip is located at the free end of a cantilever (50-200 μm long) with a spring constant of 40 to 0.01 N/m. Typical forces of a few pN between the tip and the sample surface cause vertical deflection of the cantilever on the $\text{\AA}$ scale. How can such small cantilever deflections be detected? This can be done by the so-called optical lever principle: A laser beam is directed to the back of the cantilever and is reflected to a position-sensitive detector (PSD). As the cantilever deflects the position of the laser beam on the detector shifts (Figure 6). The PSD itself can measure displacements of the laser beam as small as 1 nm. However due to the large path length (several centimeters) between cantilever and the PSD small deflections of the cantilever are translated in large changes of the laser position at the PSD (optical lever principle). As a result, the system can detect sub- $\text{\AA}$ deflections of the cantilever. In this way the small forces the tip and the sample experience lead to small deflections of the cantilever. This can be used to generate an image of the sample by scanning the tip for a defined area along the sample.

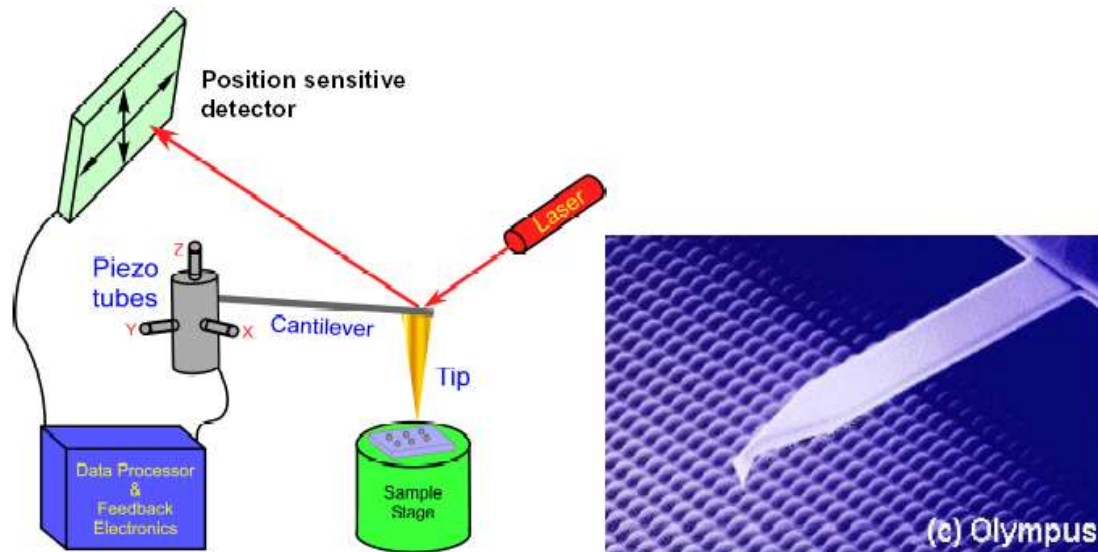


Figure 6: Principle of AFM (left), electron micrograph of an AFM cantilever above the sample surface (right).

The interaction forces in the AFM are often quite complex due to several factors:

- Even if the tip apex should be monoatomic, the number of atoms from the tip involved in the interaction is the sum over many interactions including long range forces like electrostatic interactions
- The forces are strongly dependent on the environment (gas, liquid or vacuum)
- The scan is a dynamic process, which means that the rate dependency of forces need to be considered, particularly for chemical interactions (covalent, ligand/receptor interactions)
- The tip can deform and manipulate the sample, and vice versa the sample can compromise the surface composition of the tip while scanning

Several techniques have been developed to allow precise detection of the sample structure despite of these difficulties, e.g. the so called tapping mode as described below. An exhaustive insight into this technique and results on organic and biological materials can be found in the remainder of this section.

5.2. Forces in AFM

It is important to distinguish the type of forces between the tip and the sample in order to separate the contributions and correctly interpret the experimental results.

(i) Long range forces

1. Van der Waals forces: they are due to atomic polarizability and therefore very general. They are proportional to $1/r^6$ where r is the distance between them. The role of these forces in AFM has been discussed by Moiseev [6] and Hartmann [7]. They have a spatial extent that goes from one to tens of nanometres, which is considered “long-range” in the context of AFM interaction forces.
2. Electrostatic forces: they are due to coulomb interactions; in the present case they can occur between an electrostatically charged tip and a charged area of an insulating surface. They have a spatial extent that goes up to hundreds of nanometres.
3. Capillary forces: the curvature at the contact between the tip and the sample causes the condensation of vapor from the ambient including water from the air. Also surfaces exposed to the air environment are typically coated by a layer of water, whose thickness depends on the relative humidity (RH) of the atmosphere and on the physico-chemical nature of the object. Strong attractive capillary forces (about 10^{-8} N) result, which hold the tip in contact with the surface. To avoid capillary forces the relative ambient humidity must be at RH = 0%, although Thundat and co-workers demonstrated that below RH = 10% they could not detect decays any further of the capillary forces [8]. Two simple experimental procedures can minimize the effect of this kind of forces. Firstly, working in a fluid cell with both tip and sample immersed in a liquid, or secondly, using a sealed chamber for the measurements filled with an inert gas e.g. dry nitrogen.

(ii) Short Range Forces

1. *Repulsive forces*: the interatomic repulsion forces is due to electrostatic repulsion between nuclear charges. They are proportional to $1/r^n$ with $n > 8$. The large exponent n implies that the forces are extremely short-range and level off quickly as the interacting surfaces are placed at larger separations.
2. *Forces of covalent bonds*: they originate from the overlap of the wave functions of two or more atoms. In this case the density of electronic charges is concentrated between the two nuclei. This force decreases abruptly for a separation over a few ångströms. The type of interaction can be also called *chemisorption*.
3. *Metallic adhesion*: they derive from the interaction between strongly delocalized electronic clouds, which cause strong interactions that decay exponentially with distance. They are important when two metallic surfaces approach to the extent that the electronic wave functions overlap [10]. This case can be called also *physisorption*.
4. *Friction*: during the scan, there is a force component parallel to the surface, since the tip is not always oriented exactly perpendicular to the surface. This friction tends to twist the cantilever, and since the torsion angle depends on the composition of the surface, the

measurement of the twist provides chemical information. It was also shown that this kind of friction force can be detected at the atomic scale.

Lennard-Jones potential

As a first approximation, the forces contributing to the deflection of an AFM cantilever can be considered the *van der Waals* and the *interatomic repulsive forces*, which can be derived from the Lennard-Jones potential.

$$\varepsilon_r = -4\varepsilon_0 \left[\left(\frac{\sigma}{r} \right)^6 - \left(\frac{\sigma}{r} \right)^{12} \right] \quad (1)$$

With: ε_0 = potential energy at the minimum, s = effective molecular diameter, r = between interacting surfaces distance.

The interaction force is the gradient of the interaction energy; its dependence on the distance between two interacting surfaces is depicted in Figure 7. Two distance regimes are highlighted: 1) the contact regime; and 2) the non-contact regime. In the contact regime, the cantilever is positioned such that short-range repulsive interactions prevail, and the interatomic force between the cantilever and the sample is repulsive. In the non-contact regime, the cantilever is held on the order of tens to hundreds of Ångströms from the sample surface, and the interatomic force between the cantilever and sample is attractive (largely a result of the long-range van der Waals interactions). Both contact and non-contact imaging techniques are described in detail in the following sections.

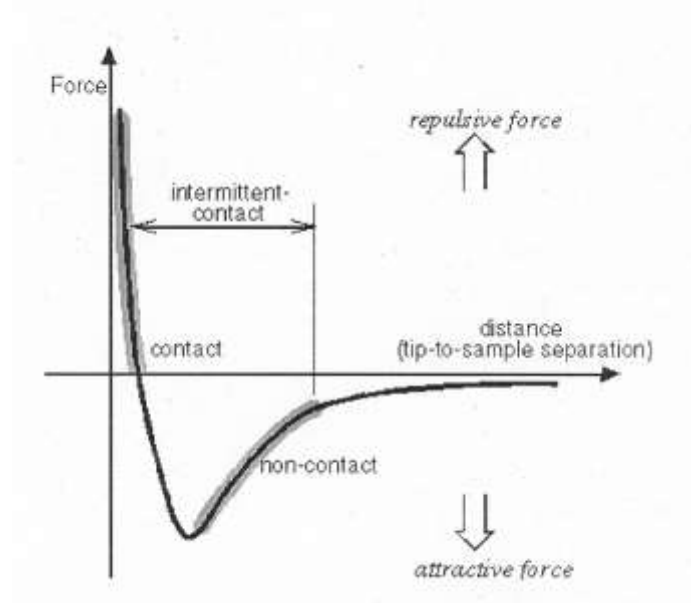


Figure 7: The Lennard-Jones potential in form of a force vs. distance curve with regions of contact, intermittent contact, non-contact mode.

5.3. Contact Mode AFM

In contact mode an AFM tip makes soft physical contact with the sample. The tip is attached to the end of a cantilever with a low spring constant, lower than the effective spring constant holding the atoms of the sample together. As the scanner gently scans the tip across the sample (or the sample under the tip), the forces experienced upon contact by the tip causes the cantilever to bend, allowing the topographic profile to be mapped. Using very stiff cantilevers it is possible to exert large forces on the sample and the sample surface is likely to be deformed; this may also be used in nanolithography. The total force that the tip exerts on the sample is the sum of the capillary plus cantilever forces, and must be balanced by the repulsive and van der Waals forces for contact AFM. The magnitude of the total force exerted on the surface varies from 10^{-8} N (with the cantilever pulling away from the sample almost as hard as the water is pulling down the tip), to the more typical operating range of 10^{-7} to 10^{-6} N. The contact mode AFM can generate the topographic data set by operating in one of two modes: constant height or constant force mode. In *constant height mode*, the spatial variation of the cantilever deflection can be used directly to provide the topographic data set because the height of the scanner (consequently also the distance sample surface - tip holder) is fixed as it scans. The far more relevant mode for AFM imaging is the *constant force mode*. Here the deflection of the cantilever is used as input to a feedback loop that moves the scanner up and down in Z-direction, responding to the topography by keeping the cantilever deflection constant, see Figure 6. In this case, the image is generated from the scanner's motion. With the cantilever deflection held constant, the total force applied to the sample is constant. In constant force mode, the speed of scanning is limited by the time of response of the feedback loop, but the total force exerted on the sample by the tip can be controlled. This mode is usually preferred for most applications because it gives a real topographic map of the sample surface. Constant height mode is often used for recording atomic-scale images of atomically flat surfaces, where the cantilever deflections and thus variations in applied force are small. This latter mode is also used for monitoring fast processes in real-time, where high scan speed is essential.

5.4. Vibrating mode AFM: Non-contact and intermittent contact

To overcome the problem of the friction component during scanning in contact mode, and to minimize the forces exerted from the tip on the sample and the effect of the capillary forces, alternative modes have been invented where the AFM cantilever vibrates near (on the order of tens to hundreds of ångströms) the surface of a sample (Figure 8). Basically, a cantilever is forced to oscillate near its resonant frequency (typically from 50 to 400 kHz) with an amplitude of a few hundreds of ångströms. While the tip scans over the sample the cantilever oscillation amplitude is monitored and kept constant with the aid of piezo feedback loop that moves the scanner up and down depending on the sample height. By keeping the amplitude constant, the system is expected to also keep the average tip-to-sample distance constant, see Figure 9. The sensitivity of this detection scheme provides sub-ångström vertical resolution in the image, as in contact AFM. Due to the elimination of the shear forces that are applied from the tip to the sample, these modes are particularly useful for studying soft or elastic materials such as biological and organic films. As a consequence of the reduction of the overall interaction forces between the tip and the sample surface, these modes do not suffer from tip or sample degradation effects as much as in contact mode AFM. Unfortunately, the lateral resolution that can be reached is a few nanometers, which is lower than in the contact mode.

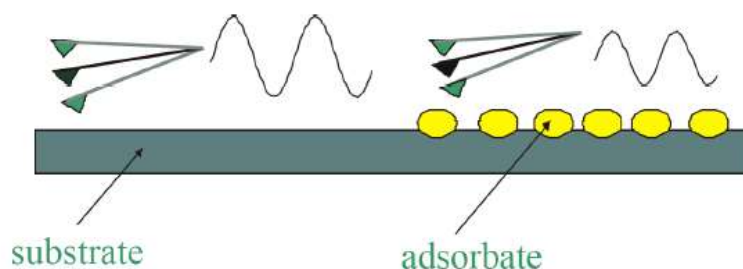


Figure 8: Vibrating modes: the tip-sample interaction decreases the amplitude and shifts the phase of oscillation.

(i) Non-contact Mode

In this mode the tip-sample interaction is indicated on the force-distance curve of Figure 2 as the non-contact regime. Since the force between the tip and the sample in this regime is low (generally about 10^{-12} N), the force measurement is more difficult than in the contact regime, where it can be several orders of magnitude larger. Furthermore, cantilevers used for non contact mode must be stiffer than those used for contact AFM because soft cantilevers can be pulled into contact with the sample surface. The small force values in the non-contact regime and the greater stiffness of the cantilevers used for NC-AFM are therefore both factors that limit the force resolution, and consequently the lateral resolution which can be achieved.

5.5. Intermittent contact mode (also termed tapping mode)

Tapping Mode or intermittent-contact AFM (TM-AFM) is similar to the non contact mode, except that for tapping mode the vibrating cantilever-tip is brought closer to the sample so that at the bottom of its travel it just barely hits, or taps the sample. The intermittent-contact operating region, indicated on the curve in Figure 7, notably increases the lateral resolution. In general, it has been found that TM-AFM is more effective than NC-AFM both for imaging larger scan sizes that may include greater variation in sample topography and for the slightly higher resolution that can be achieved due to the stronger tip-sample interaction forces that are sampled. Making use of the phase imaging mode, which operates detecting the shift in phase of the vibration, it possible to increase further the spatial resolution. In this imaging mode the contrast originates from differences in surface adhesion and viscoelasticity; it is therefore very helpful to detect different phases that coat the sample surface. The basic set-up of tapping mode AFM is shown in Figure 9.

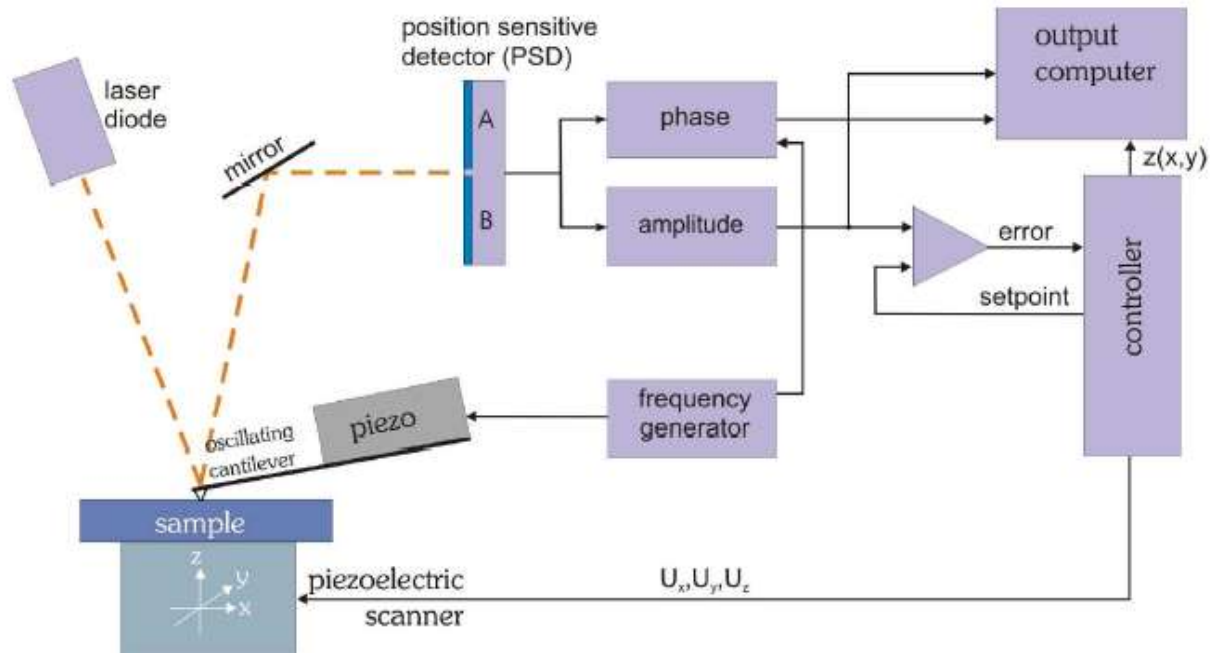


Figure 9: The scheme for tapping mode AFM.

5.6. Piezoelectric scanners and scanning artifacts

Actuators (scanner) used in AFM are usually made of a single piezoceramic tube that is radially polarized. The tube is plated on the inside and on the outside with an electrode material (usually nickel). The outer electrode is divided along the tube axis into four equal segments (Figure 10). The tube bends when a voltage is applied between one of the outer electrodes and the inner electrode, due to the piezoelectric transverse effect. When opposite segments (referred to as x and -x or y and -y) are driven by a signal of the same magnitude but opposite sign, the tube is bent twice as much. The bending permits the probe or the sample to move in two dimensions (approximately on a sphere). The inner electrode is usually driven by the signal that adjusts the sample-probe-distance (usually referred to as z-direction).

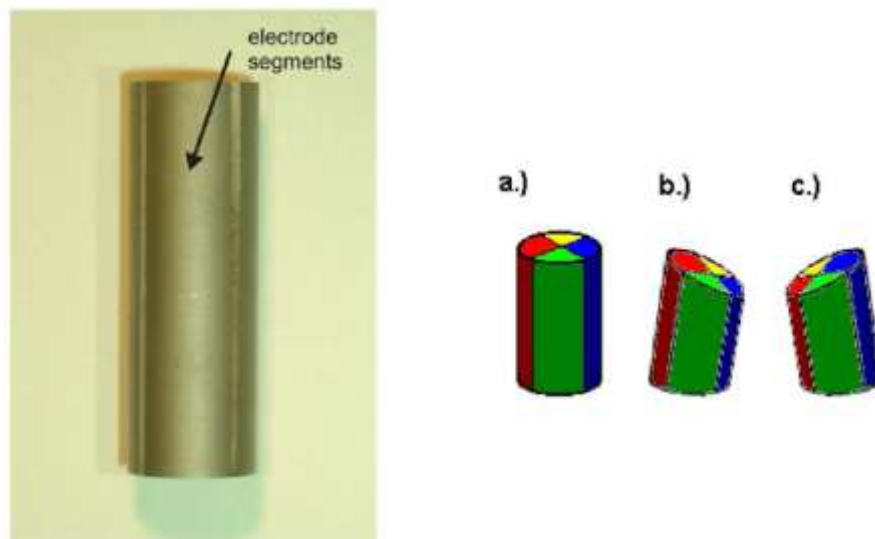


Figure 10: Piezoelectric scanner tube (left), scheme of the movement of a piezoelectric scanner tube.

The applied electric field, E , and the relative strain, S , (the relative deformation of a body) are related merely by the piezoelectric strain constant, d , when the actuator is allowed to freely deform (without mechanical stress). $S = d E$. Since d is temperature dependent and not truly constant for large applied electric fields (e.g. large driving voltages) also the sensitivities for the x, y and z direction of a piezoelectric tube are not constant. Furthermore, the piezoelectric actuators follow a hysteresis curve and the displacement of an actuator is time dependent (creep). All these problems make it difficult to calibrate a piezoelectric tube and introduce scanning artifacts when the piezo is driven over large distances.

5.7. Scanning artifacts due to tip wear and convolution

After continued scanning it is often observed, that the size of the imaged objects increases and the resolution generally decreases. Also, very small and steep objects can usually appear enlarged or with “softened” edges even with new AFM tips. This is due to the so-called convolution effect that describes an imaging artifact that appears when the curvature of the AFM tip is smaller (less pointed) than that of the sample, see Figure 11.

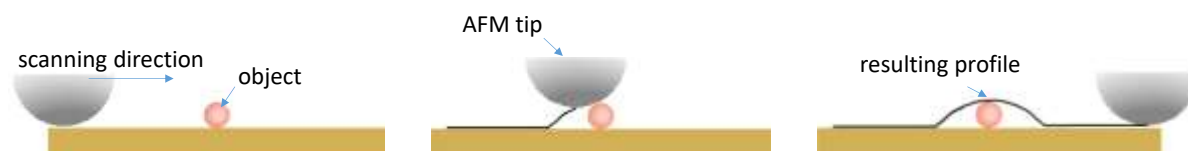


Figure 11: In this case the curvature of the AFM tip is smaller than the objects' curvature. The convolution effect takes place and so instead of imaging the sample topography, the tip topography (or parts of it) are imaged.

As a result the resolution in AFM is limited by the curvature of the AFM tip. A comparison between a standard AFM tip (10 nm radius curvature) and AFM tips modified with a carbon-nanotube (1 nm radius curvature) is shown in Figure 12.

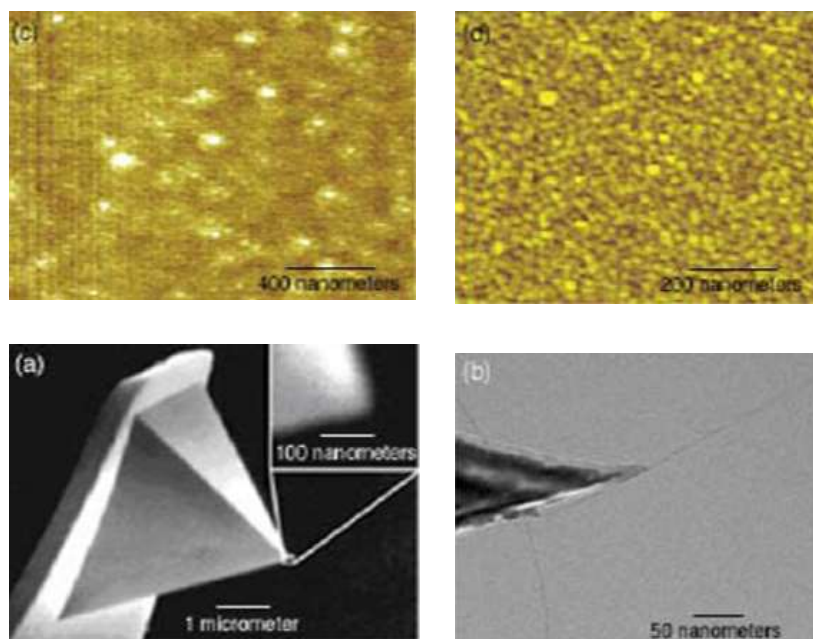


Figure 12: Left panel: image of TiO_2 nanoparticles with standard AFM tip. Individual particles are not resolved. Right panel: Image of the same sample with a carbon nanotube modified cantilever. Individual TiO_2 nanoparticles are resolved.

This problem becomes increasingly relevant for prolonged scanning further reducing the tip curvature due to wear, see Figure 13

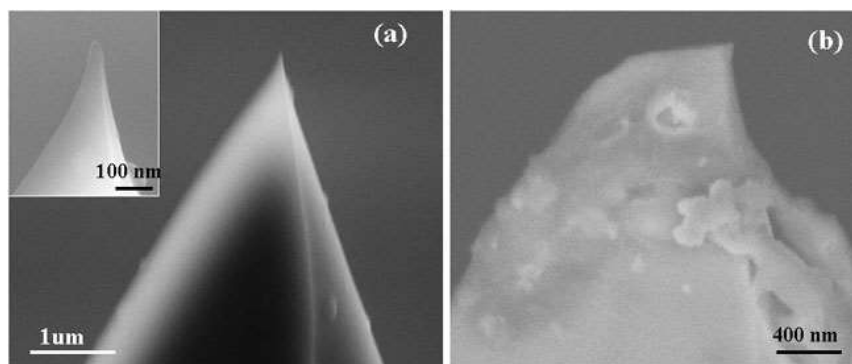


Figure 13: New tip (left); Tip after prolonged scanning (right)

6 Introduction to AFM Force spectroscopy^{7,8,9}

The AFM is best known for its high-resolution imaging capabilities, but it is also a powerful tool for sensitive force measurements. Information about the sample is also available from measuring the changes while the separation from the surface is varied at a single point, rather than by scanning the lateral position of the tip. In this mode the base of the cantilever is moved in the vertical direction towards the surface using the piezo and then retracted again. During the motion, the deflection of the cantilever and other signals, such as the amplitude or phase in dynamic AFM modes, can be measured. This is usually called force spectroscopy. The AFM tip is able to probe an extremely small interaction area (using a tip radius in the range of 5-50 nanometers), and this gives it a high sensitivity to small forces. The study of interaction forces with the AFM has led to a deeper understanding of many biological and physical processes down to the single molecule level.

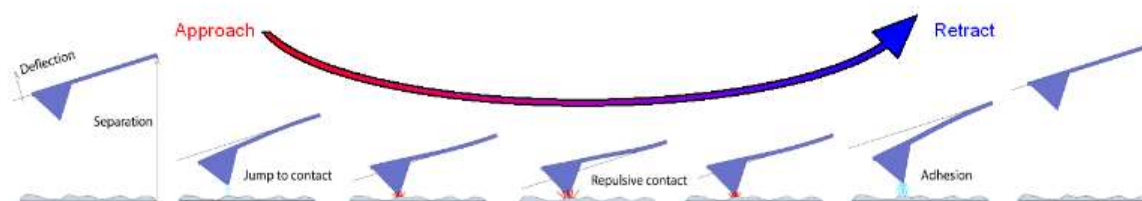


Figure 14: Schematic diagram of the vertical tip movement during the approach and retract parts of a force spectroscopy experiment (left).

⁷ (1:1 copied from JPK instruments AG handbook)

Further reading:

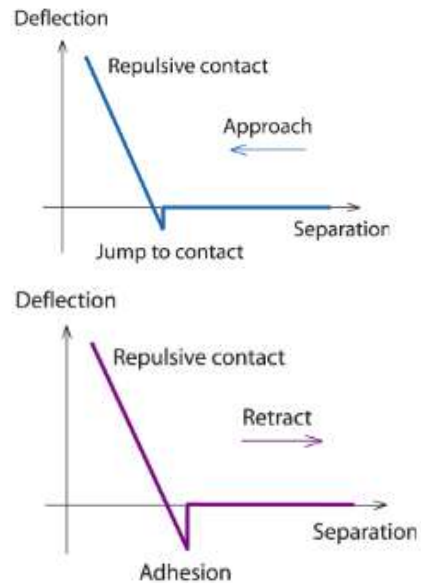
⁸ Jena, BP & Hörber, JKH (eds.) *Atomic force microscopy in cell biology*. Academic press, San Diego & London 2002. ISBN: 0-12544171-1 (hb), ISBN: 0-12383851-7 (pb)

⁹ H.-J. Butt, B. Cappella, M. Kappl; *Surface Science Reports* **59**, 1 (2005).

6.1. Simple force curves

The data from an experiment is often displayed as a simple x-y plot. The height positions for the approach or retract of the cantilever are usually chosen as the x-axis, and the cantilever property that is being measured is the y-axis. This is usually the vertical deflection of the cantilever, which can give a direct measure of the interaction force. These "force-distance" plots are often referred to as force curves. The basic force spectroscopy curves can be understood by thinking about the example of a cantilever in air approaching a hard, incompressible surface such as glass. As the cantilever approaches the surface, initially the forces are too small to give a measurable deflection of the cantilever, and the cantilever remains in its undisturbed position. At some point, the attractive forces (usually Van der Waals and capillary forces) overcome the cantilever spring constant and the tip jumps into contact with the surface. This is labeled "Jump to contact".

Once the tip is in contact with the sample, it remains on the surface as the separation between the base and the sample decreases further. This causes a deflection of the tip and an increase in the repulsive contact force. As the cantilever is retracted from the surface, often the tip remains in contact with the surface due to some adhesion and the cantilever is deflected downwards. At some point the force from the cantilever will be enough to overcome the adhesion, and the tip will break free.



6.2. More complex interactions

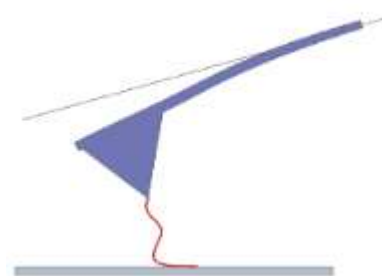
Many interesting samples are not hard and incompressible, and a more general pair of approach and retract curves will include sample compression, hysteresis and more complex adhesion between the tip and surface. In liquid, there may not be an obvious snap to contact in the approach curves even over a hard surface. Over a soft, compressible sample in liquid, the force curve often shows a gradual increase in force, without the sharp onset of the interactions seen in air. It is often difficult to define a single point where the tip and sample come into “contact”, since the initial compression of the surface causes very little deflection of the cantilever. The gradient of the repulsive contact region changes as the sample is indented and the apparent stiffness may change as the structure is compressed. For thin samples on a hard surface, the linear repulsive contact regime may be seen at large deflections, as the tip may indent the sample enough to feel the supporting surface below. The contact area will change as the tip indents a soft surface, so the actual interactions involved in compression can be hard to quantify, and different points within the region will experience different levels of compression. When the tip is retracted from the surface, there is often a hysteresis seen, if the sample is not perfectly elastic, and many different adhesion responses can be observed. In some cases, the cantilever pulls the tip free in stages, such as when there are long molecules in the sample or on the tip. Extendable contacts are made between the tip and sample, so that as the base of the cantilever retracts, the tip is deflected down towards the sample until the force is strong enough to break the contacts. Different molecules or parts of the sample may adhere and each part may be broken separately, or together. These situations produce a variety of adhesion events, and successive force curves can show very different responses. The force curves are often repeated at different locations to build up a map of the tip-surface interaction, or repeated many times at the same point to give a full statistical understanding of the interaction.

6.3. Applications

There are a huge number of potential applications of force spectroscopy, ranging from nano-mechanical investigations of elastic properties to protein unfolding and investigations of single chemical bonds, so only a brief overview is possible here. Virtually any sample can be studied using force spectroscopy, and different interactions or tip coatings and shapes will all give complementary information about the sample.

Molecular stretching

When molecules are attached to the tip and/or the sample, the stretching, unfolding or adhesion of single molecules can be studied. Long chain molecules, such as DNA or dextran can be stretched between the tip and the sample. The stiffness, persistence length and internal molecular transitions can be studied. The melting transition in DNA can be seen as the backbone rearranges under raised tension. Molecules with complex 3-dimensional structure, such as many proteins, can be unfolded in a controlled way so that the structural units can be investigated. Titin and bacteriorhodopsin are examples of proteins that have been intensively studied. Membrane proteins can be pulled out of the membrane, and the “popping” out of individual alpha-helices has been seen



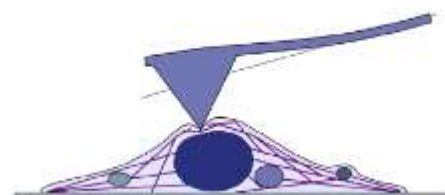
Ligand receptor interactions

The adhesion can be measured between molecules attached to the tip and to the sample. These can be antibodies and antigens or other receptor-ligand pairs. The adhesive forces can be measured and mapped over the surface, and information extracted about the energy and kinetics of the binding. These techniques have also been applied to the binding between complementary and mismatched DNA strands



Micromechanical measurements

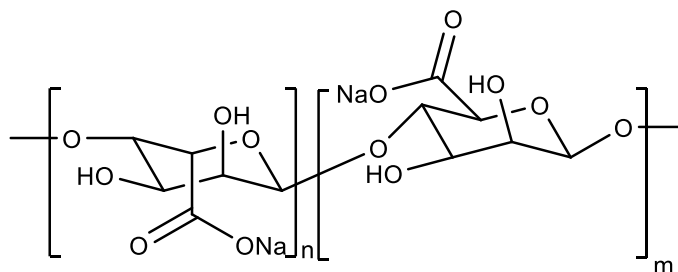
The viscoelastic response of hydrogels or cells can be studied by using the cantilever to indent the respective specimen. On living cells, the changes in mechanical properties can be seen as the cell divides, or when drugs such as cytochalasin, which disrupts the cytoskeleton, are added. Mechanosensitive cells such as osteoblasts or ear cells can be stimulated with the cantilever, and the response monitored. Adhesion maps over the surface are also possible to investigate the distribution of receptors.



7 Practical Course

- 7.1. Synthesis of Physically Cross-linked Gel-Building Polymer
- 7.2 Synthesis of Chemically Cross-linked Gel-Building Polymer
- 7.3 Synthesis of poly-(N-isopropylacrylamide) microgels via precipitation polymerization.

7.1. Synthesis of non-Covalently Cross-linked Polymer Gel¹⁰



Of all the different biopolymers that can be employed in the formation of the semipermeable membrane in capsules, alginate is one of the most frequently used owing to the fact that its immobilization is carried out under very mild conditions. In molecular terms, alginate consists of a family of unbranched binary copolymers of (1-4)-linked β -D-mannuronic acid (M) and α -D-guluronic acid (G) of widely varying composition and sequential structure. Alginates are true block copolymers composed of homopolymeric regions of M and G, termed MM and GG blocks, respectively, interspersed with regions of alternating structure (MG blocks). In solution, alginates behave like flexible coils. However, upon interaction with divalent metal ions, such as Ca^{2+} , they form an ordered structure. Grant et al. showed that the gelation process involves cooperative binding of calcium ions between aligned GG blocks of two alginate chains. The gelation of calcium alginate is an almost instantaneous and irreversible process, which is governed by the relative rate of diffusion of calcium ions and polymer molecules into the gelling zone. This fact indicates that the gelation process can be accurately expressed by the relationships used for other diffusion-limited reaction systems.

Materials

Sodium alginate
Calcium chloride
Distilled water

Water bath
Beaker flask (3 x 100 mL)
Büchner funnel
Filter paper
Magnetic stirrer (with magnetic stir bar)
Magnetic stirrer (with contact thermometer and magnetic stir bar)
Syringe (2 x 2,5 mL)

¹⁰ Ana Blandino, Manuel Macias, Domingo Cantero, Formation of Calcium Alginate Gel Capsules: Influence of Sodium Alginate and CaCl_2 Concentration on Gelation Kinetics, Journal of Bioscience and Bioengineering (1999), Vol. 88, No. 6, 686-689.

1) In a water bath, a beaker flask with 50 mL water is heated at about 90°C. Then sodium alginate (200 mg) is dissolved in the hot water.

2) In a second and third beaker flask, calcium chloride is dissolved in water.

3) **Alginate particles:** The sodium alginate solution is dropped (about 10 drops) through a syringe into the calcium chloride (50 mL, 0,5 M). It can be observed, that sodium alginate builds a white precipitate (globular particles) in calcium chloride solution. These are taken out and the water is removed with a syringe.

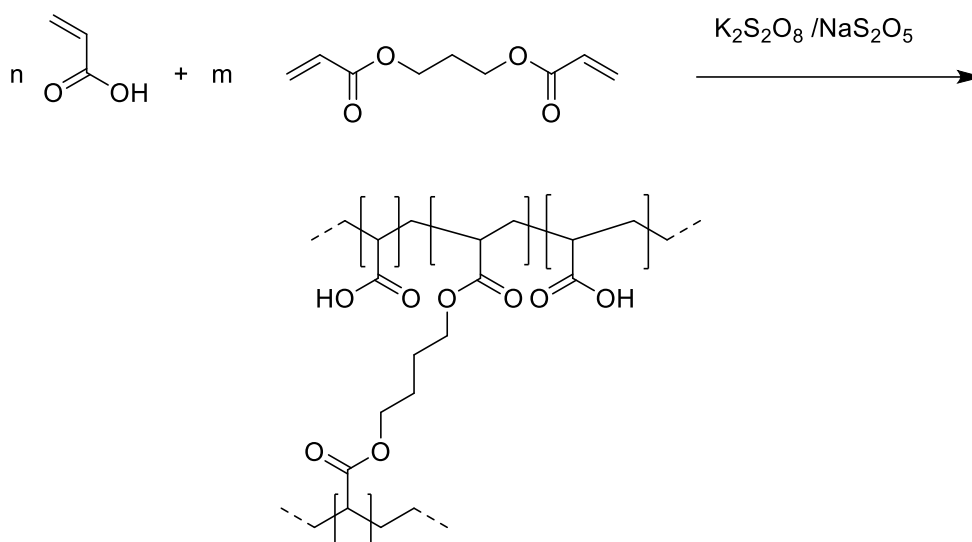
4) **Alginate capsules:** Droplets of calcium chloride solution (50 mL, 1 M) are added dropwise (about 5 drops) through a tube into sodium alginate solution, using a syringe. A dropping height of about 10 cm ensures that spherical droplets are formed. The sodium alginate solution is constantly stirred at 350 rpm using a magnetic stirrer situated at the bottom of the vessel, in order to keep the droplets from sticking together and to minimize external transfer resistance.

Once the CaCl_2 solution was added dropwise into the alginate solution, a capsular membrane is formed instantaneously around each droplet due to the cross-linking of the interfacial alginate molecules with calcium cations.

Alginate capsules and particles are deposited onto a plastic petri dish in order to confirm their different morphology.

Photographs are taken of both (particles and capsules).

7.2. Synthesis of Chemically Cross-linked Gel-Building Polymer¹¹



Materials

Acrylic acid	7,2 g (0,1 mol)
Butandiol diacrylate (BDDA)	200 mg
Sodium disulfite	0,5 mol%, referring to acrylic acid
Potassium peroxodisulfate	0,5 mol%, referring to acrylic acid
Ultra-pure water	
Sodium chloride solution	100 ml (0,5 M)
Calcium chloride solution	100 ml (0,17 M)

Tri-necked flask (100 mL)
 Beaker flask (3 x 100 mL)
 Reflux condenser
 Filter paper
 Bubble counters
 Magnetic stirrer
 Egg shaped magnetic stir bar
 Glass stick
 3x Petri dish
 3x Watch glass
 Glass stopper NS 14,5
 Silicon stopper
 1x Tweezers

¹¹ D. Braun, H. Cherdron, M. Rehahn, H. Ritter, B. Voigt. Polymer Synthesis: Theory and Practice, 5th Edition, Springer 2012, p. 378.

Synthesis:

Acrylic acid and 20 mL of distilled water are placed in a 100 mL flask and stirred for 15 min under nitrogen. Butandiol diacrylate (BDDA) is added to the mixture. 0,5 mol% of sodium disulfite and 0,5 mol% of potassium peroxodisulfate is dissolved in 1 mL water and added to the solution. The reaction mixture is stirred 1 h under nitrogen. The gel effect occurs after 1 h and the polymerization ends.

Ionic strength dependent swelling:

Synthesized hydrogels are broken into small pieces and dried under vacuum (80 °C, 2 hours). To characterize the superabsorbent polymer, first a teabag is added into distilled water. After a few minutes the teabag is removed and allowed to drain for ten minutes. The weight of the drained teabag is determined. Subsequently, three teabags with 2 g of the superabsorbent polymer are prepared. One is put into 100 mL ultrapure water the second one in 100 mL NaCl-solution (0,5 M) and the third in 100 mL of a 0,17 M CaCl₂-solution. The water absorption is determined by weighing the drained teabag after one, (two, three) hours and on the next day (about 20 hours) using a balance..By the use of the followed equation the swelling degree can be defined:

$$q = \frac{(a - b)}{b}$$

q= degree of swelling, a= mass of swollen network, b= mass of dry mass

7.3. Synthesis of poly-(N-isopropylacrylamide) microgels via precipitation polymerization

a) Studying the impact of initiator variation

Materials

N-Isopropylacrylamide (NIPAM)	0.7 g (6.2 mmol)
N,N-Methylene(bis)acrylamide (MBA)	0.05 g (0.325 mmol)
Sodium dodecyl sulfate (SDS)	0.025 g (0.087 mmol)
Ammonium persulfate (APS)	Different amounts*
Ultrapure water	

- * Group A: 0.1 g (0.438 mmol)
Group B: 0.5 g (2.191 mmol)

b) Studying the impact of surfactant variation

Materials

N-Isopropylacrylamide (NIPAM)	0.7 g (6.2 mmol)
N,N-Methylene(bis)acrylamide (MBA)	0.05 g (0.325 mmol)
Sodium dodecyl sulfate (SDS)	Different amounts*
Ammonium persulfate (APS)	0.2 g (0.876 mmol)
Ultrapure water	

- * Group A: 0.025 g (0.087 mmol)
Group B: 0.050 g (0.174 mmol)

Tri-necked flask (250 mL)
Bubble counters
Reflux condenser
Glass stopper
Silicon stopper
Magnetic stirrer
Egg shaped magnetic stir bar
Tapping mode cantilever

NIPAM, MBA and SDS are dissolved in 150 mL ultra-pure water. The mixture is then heated to 70°C under constant nitrogen flow for 40 minutes. The initiator (APS) is meanwhile dissolved in 10 mL ultra-pure water and injected with a syringe into the solution. The solution will start to get a blueish blur. The mixture is stirred (200 rpm) for 2 hours under constant nitrogen flow. After polymerization the solution is left to cool.

Dynamic light scattering: Next, 200 µL of the microgel dispersion are deposited into a plastic cuvette and filled up with 2 mL of water to perform dynamic light scattering experiment. The particle size is determined via dynamic light scattering at 20°C and 40°C as instructed by the supervisor. Note the temperature dependent swelling of the particles and their overall size in solution.

AFM imaging: The coverslips are placed on a microscope slide using a double adhesive tape for fixation. The coverslip should then be imaged by AFM using cantilevers with a spring constant of 10-40 N/m in intermittent contact mode. Scan size circa 10 μm . Note the near monodisperse shape of the microgels as well as their (possibly) regular packing on the coverslip.